

Write a program using C/C++/Java to simulate the first fit, best fit and worst fit memory allocation strategy. Assume memory chunk and initial requirement for memory block from your side.

Example Program Flow:

1. Initialize the program with an initial memory chunk of a specified size.
2. Display the menu for the user to choose an allocation strategy (First Fit, Best Fit, Worst Fit) or to exit.
3. Prompt the user to request a memory block allocation, specifying the size.
4. Allocate memory based on the chosen strategy.
5. Display the updated state of the memory chunk.
6. Repeat steps 3 to 5 until the user chooses to exit.

```
ubuntu@ubuntu-VirtualBox:/media/sf_Ubuntu_Files/Lab8$ ./Lab8
Enter the total memory size: 1000

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit
Enter your choice: 1
Enter the size of memory to allocate: 200
Memory block of size 200 allocated at address -2096842744.

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit
Enter your choice: 2
Enter the size of memory to allocate: 300
Memory block of size 300 allocated at address -1707619329.

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit
Enter your choice: 3
Enter the size of memory to allocate: 150
Memory block of size 150 allocated at address 500.
```

Memory block of size 150 allocated at address 500.

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit

Enter your choice: 5

Memory Status:

Start Address: 0, Size: 200, Allocated

Start Address: 200, Size: 300, Allocated

Start Address: 500, Size: 150, Allocated

Start Address: 650, Size: 350, Free

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit

Enter your choice: 4

Enter the start address of the block to free: 200

Block at address 200 has been freed.

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit

Enter your choice: 2

Enter the size of memory to allocate: 100

Memory block of size 100 allocated at address -1707619329.

Memory block of size 100 allocated at address -1707619329.

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit

Enter your choice: 5

Memory Status:

Start Address: 0, Size: 200, Allocated

Start Address: 200, Size: 100, Allocated

Start Address: 300, Size: 200, Free

Start Address: 500, Size: 150, Allocated

Start Address: 650, Size: 350, Free

1. First Fit Allocation
2. Best Fit Allocation
3. Worst Fit Allocation
4. Free Memory Block
5. Display Memory Status
6. Exit

Enter your choice: 6

Exiting...